Slocum Ballasting Procedure

Ballasting adjusts the glider's mass and mass distribution so it is neutrally buoyant and correctly trimmed in the water it will actually be deployed into — not the tank it's ballasted in. Get it wrong and the glider flies asymmetric dive/climb profiles, wastes buoyancy-pump energy fighting an unwanted heaviness or lightness, or in a bad case just sits on the bottom or can't submerge.

Source

Paraphrased from the *Slocum G3 Glider Maintenance Manual* (Chapter 3, "Ballasting"), the *Slocum G3 Glider Operators Manual*, and the Teledyne Webb Research operator training course. Always defer to the official Teledyne documentation, and use TWR's **Ballast Adjustment Spreadsheet** (contact `glidersupport@teledyne.com` for a copy) for the actual tank-to-target weight calculation rather than hand-deriving it.

Before you start: vacuum

The glider must be closed with a proper vacuum **before** it is powered up — see the one-pager. Evacuate to at least **6 in Hg (shallow glider)** or **7 in Hg (deep glider)**; pulling a higher vacuum than the minimum is better practice, since some air bleeds in once the glider is powered. Seal with the 7/16-20 PEEK MS plug torqued to **15 in-lb** — PEEK is delicate, so use the correct tools and torque.

The goal

Ballasting is done with the glider in a defined reference state: